A Data-driven Approach to Profiling Coping Behaviors during the COVID-19 Pandemic Using Automated Clustering Algorithms

A Research Project by

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This research paper, entitled "A Data-driven Approach to Profiling Coping Behaviors during the COVID-19 Pandemic Using Automated Clustering Algorithms", prepared and submitted by Kristine Bernadette Q. Nunez, in partial fulfillment of the requirements for the course CS 199 is hereby accepted.

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Abstract

Restrictions on mobility and physical contact implemented to contain the COVID-19 pandemic resulted in mental health issues. Studies indicated that the coronavirus outbreak caused an increase in stress, anxiety, and Post-Traumatic Syndrome Disease (PTSD) in affected populations, the impact of which depended on individual coping strategies. Due to the possibility of another pandemic, it is important to understand coping mechanisms to formulate intervention methods and public health policies.

This study aimed to develop a machine learning model of coping profiles of the Japanese population based on an online survey conducted from May 13 to 30, 2022, when the number of COVID-19-positive cases was high. Using the K-Means clustering algorithm and the elbow method, 5 clusters were identified from the socio-demographic, mental health status, coping behavior, and COVID-related profile data of 16,641 respondents. The 10 features that define each cluster are loneliness, well-being, psychological stress, post-traumatic growth (PTG), amount of exercise, continuous prevention, trust in government, optimism, interaction deterioration, and economic deterioration. The clusters can be generally described as semi-optimistic/semi-pessimistic (Cluster 1), optimistic (Cluster 2), resilient (Cluster 3), pessimistic (Cluster 4), and highly-stressed (Cluster 5). PTG was observed to be a determining factor that drives the characteristics of each cluster. Clusters 2 and 3, despite having high PTG scores, comparatively differed in well-being and stress conditions, highlighting the significance of coping mechanisms. This supports past research findings that post-traumatic growth is experienced by both resilient and stressed individuals, using different strategies to cope. These results can be used as a reference for behavioral

management of population segments to minimize the mental health impact of a pandemic.

Acknowledgement

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I hope that when we look back to this project,
we can only see good memories.

My deepest gratitude.

Chapter 1

Introduction

The COVID-19 pandemic drastically changed the way people live because of restrictions on movement [1] and physical contact [2] [3] implemented to contain the spread of the virus. These containment methods resulted to isolation and caused mental health issues across countries [4] [5] [6]. A survey in Japan conducted during a mild lockdown found that 48.1% of the respondents were psychologically distressed, 1.7 times higher compared to the psychological distress reported in 2013 after the 2011 Great East Japan Earthquake [4]. In Canada, 50% of teenagers aged 13 to 19 years old experienced socio-spatial and temporal disconnections, and 31% took the emotional toll of the pandemic in form of break downs, feeling of being at a loss, and cycle of emotions. A few (0.68%) had suicidal thoughts [7]. Data taken from April to May, 2020, when Pakistan enforced a complete lockdown showed that 34% of college students had moderate to severe anxiety, and 45% had depression [8]. A similar study among 14 to 35 year-olds in China indicated that that 40.4% had psychological problems, and 14.4% experienced PTSD [9]. In a comparative study of the impacts of the COVID-19 pandemic on the mental health of young adults in Ghana, Philippines, Egypt, Pakistan, and India, the Philippines scored highest in the depression, anxiety, and stress scales [6].

Variation in response to psychological stress. Although the pandemic generally has a negative impact on mental health, certain groups of people responded differently. Whereas depression, anxiety, and PTSD were found to have no correlation with age in the United States [5], in China, those in the age group 12 to 21.4 years had significantly higher PTSD scores compared to the age group 49.6 to 59 years [10].

Genderwise, American males and females had similar PTSD experiences. However, transgenders were found to be more likely to have high PTSD levels [5]. In contrast, Chinese males experienced PTSD more than Chinese females [9]. Racial disparity was also observed on stress response of Americans. Asian Americans are less likely to have depression and PTSD compared to white Americans, and Asian Americans and Hispanics/Latinos are less likely to have anxiety [5].

Personal relations also influenced people's response to the pandemic. In China, mental health and PTSD scores are significantly higher for divorced and widowed individuals compared to single and those married or cohabiting [9], while those staying with more than 3 persons also had significantly higher PTSD scores compared to those living alone [10].

In contrast, [7] reported that 19% of Canadian teenagers experienced positive effects of the pandemic, feeling less stress and pressure, having opportunities for self-reflection, growth, and care, and having available time to connect to friends and loved ones.

Mental health and coping. Among 18- to 30-year-old Americans, high levels of loneliness and worry, and low distress tolerance corresponded to higher depression, anxiety, and PTSD scores, indicating low ability to cope with mental stress. Those with high resilience tended to have low depression and anxiety scores. Family support decreased the likelihood of depression [5].

Application of data science in infectious diseases. The COVID-19 pandemic highlighted inadequacies of the public health system with respect to speed of response, data collection, reliability, and security, evidence-based policy-making, contact tracing, and identification of effective medicine. The speed of viral transmission and the number of infected people resulted to a large, complex set of data. Data science has proven itself helpful in generating actionable insights using the available data which otherwise requires considerable resources to process.

1.1 Statement of the Problem

The foregoing data indicate that the coping mechanisms of individuals depend on sociodemographic, psychological, and social factors, meaning, two individuals may react differently to the same stressor. Considering the general population, the interaction of these factors determines how certain groups of people cope with a pandemic or a disaster. Thus, the coping profile of a population must be studied in order to draw targeted interventions during pandemics or disasters.

1.2 Significance of the Study

Due to the uncertainty of COVID-19 and the possibility of a similar pandemic, it becomes important to understand coping mechanisms [11] [12]. Awareness about how certain segments of the population react to psychological stress triggered by the pandemic can potentially help the government in crafting public health policies, and other institutions in establishing intervention methods [13] [14]. The output of this undertaking can be used as a reference for behavioral management of population clusters during a pandemic.

1.3 Scope of the Study

This study aims to utilize cluster-based machine learning method to develop a model for COVID-19 coping profiles of the Japanese population. It covers and is delimited by the following:

(1) The data used for modelling was taken from a survey conducted in Japan [4], the design and collection of which, this study has no control of.

(2) Only DBSCAN, K-Means, and agglomerative modelling algorithms were implemented to cluster the data.

(3) The silhouette score of the resulting model was relatively low, hence, further study is needed to computationally validate the model.

Chapter 2

Theoretical Framework

2.1 Review of Related Literature

2.1.1 Coping Mechanisms and Impact of the COVID-19 Pandemic

Factors Associated with Depression, Anxiety, and PTSD Symptomatology during the COVID-19 Pandemic: Clinical Implications for U.S. Young Adult Mental Health (Liu et al., 2020) [5]

The determinant factors of depression, anxiety, and PTSD symptoms in young American adults in relation to the COVID-19 pandemic were investigated. Research targets were 18- to 30-year-old current U.S. residents or students, recruited online through organizations in New England, Midwest, South, and West regions. Survey was conducted from April 14 to May 19, 2020, one month after state of health emergency was declared, and before lifting of restrictions in the U.S.

Measurements involved (1) the 10-item Connor-Davidson Resilience Scale and the Distress Tolerance Scale for coping ability, (2) the Multidimensional Scale of Perceived Social Support for social support, (3) the 3-item version of the UCLA Loneliness Scale Short Form for loneliness, (4) the 8-item version of the Patient Health Questionnaire for depression, (5) the Generalized Anxiety Disorder Scale for anxiety, and (6) the

PTSD Checklist—Civilian Version for PTSD manifestations. A newly developed instrument was used to assess the severity of worry related to COVID-19 pandemic.

In general, the surveyed population had high loneliness (61.5% of the respondents), low resilience (72%), and low tolerance to distress (74.1%), but had good social support. High levels of depression, anxiety, and PTSD were reported by 43.3%, 45.4%, and 31.8% of the respondents, respectively.

High loneliness and worry levels and low distress tolerance corresponded to higher depression, anxiety, and PTSD scores. Those with high resilience tended to have low depression and anxiety scores. Family support decreased the likelihood of depression.

Depression, anxiety, and PTSD were found to have no correlation with age and income. Although males and females performed similarly on PTSD, high PTSD levels were more likely in transgenders. Compared to Americans, Asian Americans were less likely to have depression and PTSD; and Asian Americans and Hispanics/Latinos were less likely to have anxiety.

The Effect of COVID-19 on Youth Mental Health (Liang et al., 2020) [9]

The researchers evaluated the mental health condition and the determinant factors of mental health of the Chinese youth at the onset of COVID-19 outbreak.

A survey was conducted through WeChat, starting January 30, 2020, after COVID-19 was officially recognized by the World Health Organization. Participation was voluntary, targeting 14 to 35 years old respondents. Actual number of respondents was 584.

The knowledge of the respondents about COVID-19 was measured using a 5-item questionnaire. Assessment of mental health condition and PTSD utilized the General

Health Questionnaire (GHQ-12) and the PTSD Checklist-Civilian Version, respectively. An instrument specifically developed for Chinese respondents, the Simplified Coping Style Questionnaire, was used for measurement of negative coping styles.

SPSS 24.0 was used for statistical analysis. Associations between sociodemographic parameters, mental health condition, PTSD, and negative coping style were determined using one-way ANOVA and LSD post hoc comparisons. To study the factors affecting mental health, univariate logistic regression was utilized.

The data of 584 respondents showed that 40.4% had psychological problems, and 14.4% experienced PTSD. Psychological problems were associated with having a junior high school education or below, being an enterprise employee, PTSD symptoms, and negative coping. Mental health and PTSD scores were significantly higher for the divorced/widowed compared to single and married/cohabiting individuals, the higher scores indicating worse mental health condition. Similarly, mental health, PTSD, and negative coping scores of males were significantly higher than the females.

Generally, PTSD symptoms and negative coping mechanisms were strongly associated with poor mental health condition.

A Longitudinal Study on the Mental Health of General Population during the COVID-19 Epidemic in China (Wang et al., 2020) [10]

The study compared the mental health condition of the general Chinese population at two periods of the COVID-19 pandemic.

The first data set of the longitudinal study was taken from January 31 to February 2, 2020, during the initial outbreak of COVID-19 in China. The second data set was from February 28 to March 1, 2020, at the peak of the epidemic. A survey was conducted online across 190 cities, targeting the general population. There was a total of 1,738 respondents, 1,210 participating in the first survey, 861 in the second survey, and 333 in both.

The Impact of Event Scale-Revised (IES-R), and the Anxiety and the Stress Scale (DASS-21) were utilized for PTSD and mental health measurements. Gathering of demographic data, knowledge and concerns about COVID-19, and past 14 days history of physical symptoms, contact with COVID-19 patient, and precautionary measures was by the National University of Singapore COVID-19 questionnaire.

SPSS 21.0 was employed to conduct statistical analysis. Independent sample t-test was used to compare the psychological, depression, anxiety, and stress data from the first and second survey. Comparison of categorical data from the two surveys was by chi-squared test. To calculate associations between variables, linear regression was utilized.

Overall, the respondents of the first and second surveys had the same levels of stress, anxiety, and depression, but PTSD score was significantly higher in the first survey than in the second. For the respondents who answered both surveys, however, there was no significant difference in PTSD scores.

In the second survey, those in the age group 12 to 21.4 years had significantly higher PTSD scores compared to the age group 49.6 to 59 years. From the same survey, it was found that those staying with more than 3 persons also had significantly higher PTSD scores compared to those living alone. These observations were not present in the first survey data.

For both the first and second survey, higher levels of stress, anxiety, depression, and PTSD were directly associated with physical symptoms, very poor rating of health, and history of chronic illness. The second survey differed from the first survey in that presence of COVID-19 symptoms and recent quarantine of respondents significantly increased stress, anxiety, and depression scores.

Respondents of surveys 1 and 2 had the same view about transmission of COVID-19 by air and through contaminated objects, but significantly higher percentage of the respondents of survey 2 were uncertain about transmission by droplets. Higher proportion of survey 2 respondents also had high confidence levels about being diagnosed and surviving the virus, and were satisfied with COVID-19 information dissemination compared to survey 1 respondents. For both surveys, confidence in diagnosis, survival, and health information was associated with lower stress, anxiety, depression, and PTSD scores. However, higher anxiety and depression scores were associated with dissemination of COVID-19 health information through the radio. Moreover, adherence to COVID-19 precautionary measures resulted to lower stress and PTSD scores.

The Psychological Impact of the COVID-19 Epidemic on College Students in China (Cao et al., 2020) [15]

The study evaluated the mental health condition of college students in China during the COVID-19 pandemic, with the goal of establishing a basis for psychological interventions for college students, and for government policymaking.

The study was participated in by 7,143 undergraduate students of Changzhi Medical College in China during the COVID-19 outbreak in early 2020, while infections were uncontrolled and strict isolation measures were in place. Respondents were chosen by cluster sampling. Information gathered included demographics, source of family income, knowledge about and preventive actions related to COVID-19, and social support. Anxiety levels were measured using the Generalized Anxiety Disorder Scale (GAD-7).

Data analysis was conducted using SPSS Version 22.0. Associations between respondent characteristics and anxiety levels were determined using univariate analysis, and multivariate regression analysis was performed on statistically significant variables.

Results showed that 21.3% of students had mild anxiety, with 0.9% having severe anxiety. Factors associated with increase in anxiety were being alone, living in rural areas, unsteady family income, and COVID-19 infection of a relative or an acquaintance. Moreover, the economic impact of the pandemic, its effect on daily life, and concern about academic delays were found to be positively correlated with stress of students. Availability of social support had negative correlation with being anxious.

Psychological Impairment and Coping Strategies During the COVID-19 Pandemic Among Students in Pakistan: A Cross-Sectional Analysis (Salman et al., 2020) [8]

The research was an assessment of the psychological impact of COVID-19 on university students in Pakistan, and their coping strategies.

Data was taken from April to May, 2020, when Pakistan enforced a complete lockdown. Research participants were students of University of the Punjab, The University of Lahore, Gulab Devi Educational Complex, and University of Veterinary and Animal Sciences. There were 1,134 respondents, with a mean age of 21.7±3.5 years.

The questionnaires comprised of the Generalized Anxiety Disorder-7 for anxiety, Patient Health Questionnaire-9 for depression, and Brief-COPE for coping strategies. The survey was conducted online, using Google Forms.

Statistical analysis was conducted using SPSS 22.0. Group-to-group comparisons were implemented using independent sample t-test. One-way ANOVA was used for 3 or more groups. Significant difference among intergroup variables was tested using Tukey's HSD and Games-Howell post hoc, depending on applicability.

Results showed that 34% of the respondents had moderate to severe anxiety, and 45% had depression. Those in the age group $\geq$31 years scored lower in the depression scale compared to those in the age group $\leq$20 years. Genderwise, males were found to have less anxiety and less depressed compared to females. Higher anxiety scores were observed among respondents who had COVID-19-infected family members, friends, or acquaintances.

The most practiced coping mechanisms were religious/spiritual coping, acceptance, self-distraction, and active coping. Those in the age group $\leq$20 years were less able to cope.

The Mental Well-Being and Coping Strategies of Canadian Adolescents during the COVID-19 Pandemic: A Qualitative, Cross-Sectional Study (Ferguson et al., 2021) [7]

The paper reports the result of a study on mental health and coping mechanisms of teenagers aged 13 to 19 years old during the first wave of COVID-19 pandemic in Canada. The average age of respondents was 15.6 years.

Participants of the study, conducted from June to September 2020, were asked to answer two open-ended questions: (1) What feelings and emotions have you experienced around the pandemic? and (2) What coping strategies have you used during the pandemic? Based on the summative content analysis of all valid responses, 50% of the respondents experienced socio-spatial and temporal disconnections, 31% took the emotional toll of the pandemic, and 19% felt the positives of the pandemic.

Majority of the reported disconnections were caused by the feeling of physical isolation and missing in-person interactions, and distress about the future. The emotional toll took the form of break downs and feeling of being at a loss, and cycle of emotions. A few (0.68%) had suicidal thoughts.

The positive impacts of the pandemic were the feeling of ease due to less stress and pressure; the opportunities for self-reflection, growth, and care; and the availability of time to connect to friends and loved ones.

To cope with the physical restrictions during the pandemic, 58% of the respondents opted to do leisure and health-promoting activities, and 42% connected online and spent time outdoors.

The negative effect of restricting movements on the mental health of 81% of the teenage respondents highlights the necessity of interventions during pandemics. Such information, together with the derived information on coping strategies, can serve as bases of public health policies.

2.1.2 Clustering Algorithms

A Comparative Study of Clustering Algorithms (Gupta et al., 2019) [16]

The paper compared some commonly used clustering algorithms based on data type applicability, cluster shape, derived clustering result, sensitivity to noise, scalability, and time complexity. The algorithms were grouped according to the method generating clusters: hierarchical, partition-based, density-based, grid-based, and fuzzy-based. Comparison was made within each group.

Among the hierarchical-based algorithms (SLINK, CLINK, BIRCH, CURE, ROCK, Chameleon, DIANA, and DISMEA), only DIANA and DISMEA are not sensitive to noise. SLINK is the most noise-sensitive. In terms of scalability, BIRCH, CURE, and

DISMEA perform the best in large data sets; CLINK and BIRCH are worst for high dimensionality. BIRCH has the best time complexity.

For the partition-based algorithms (K-means, K-medoids, PAM, CLARA, and CLARANS), K-means is more sensitive to noise, it is scalable for large data sets and has the best time complexity. CLARA is also scalable for large data sets. All methods are non-scalable for high dimensionality.

All the density-based algorithms (DBSCAN, OPTICS, Mean-shift, DENCLUE, and RDBC) are not sensitive to noise, except for DENCLUE which has low noise sensitivity. Only Mean-shift is not scalable for large data sets.

Of the grid-based algorithms (STING, CLIQUE, OptiGrid, GRIDCLUS, GDILC, and WaveCluster), CLIQUE is the only non-scalable for large data sets and has the worst time complexity. OptiGrid and GDILC are not noise-sensitive, while both GDILC and WaveCluster are not scalable for high dimensionality.

Of the algorithms based on fuzzy method (Fuzzy k-means, Fuzzy k-modes, FCM, FCS, MM, and MEC), only Fuzzy k-modes is not sensitive to noise.

The comparative analysis of clustering algorithms presented in this paper can serve as a guide for identifying algorithms to be used, depending on which parameter is most important for a given application.

Hierarchical Clustering Algorithms in Data Mining (Abdullah et al., 2015) [17]

The study covered the development of hierarchical clustering method, and compared the algorithms CURE, ROCK, CHAMELEON, and BIRCH based on a review of papers from 1998 to 2015. CURE is said to be able to identify irregular shaped clusters and can isolate outliers. ROCK, on the other hand, is scalable. Although all four algorithms can handle large data sets, CHAMELEON is the most suitable for large-data applications.

Empirical Analysis of Data Clustering Algorithms (Nerurkar et al., 2017) [18]

The paper evaluated the ability of clustering algorithms K-Means, K-Means++, Kernel K-Means, Fuzzy C-Means, Model-based, DBSCAN, and OPTICS to identify clusters in the benchmark data sets CURE-T2-4K and CLUTO-T8-8K.

Only DBSCAN and OPTICS were able to detect asymmetric clusters. Although the other algorithms were able to identify the shape of asymmetric clusters, sub-clusters were defined resulting to identification of more clusters than the actual number of clusters. Overall, OPTICS performed the best in cluster detection.

Partition-based algorithms (K-Means, K-Means++, and Kernel K-Means) detected axes-symmetric clusters better than irregular-shaped clusters. Of the algorithms tested, Fuzzy C-Means was most sensitive to noise, and tended to create symmetric clusters.

Based on the data, OPTICS and DBSCAN have the best ability to detect clusters. They have worse time complexity, though, compared to the other algorithms, so the importance of this parameter would have to be factored in when choosing OPTICS and DBSCAN for a particular application. Since the experiment showed that density-based algorithms generally perform better in cluster detection, other density-based algorithms may also be considered for processing data where clustering is required.

A Density-based Algorithm for Discovering Clusters in Large Spatial Databases with Noise (Ester et al., 1996) [19]

DBSCAN and CLARANS were compared using the benchmark data SEQUOIA 2000. DBSCAN was found to be more effective in identifying irregular-shaped clusters and more efficient than CLARANS.

The study highlighted three issues commonly raised when clustering algorithms are applied to spatial data: the necessity of knowing the basic characteristic of the data in order to be able to define the input parameters, the effectiveness of the algorithm to identify clusters of irregular shape, and the efficiency of the algorithm as the data becomes larger. DBSCAN was considered for the study because it runs on a single input parameter and guides the user in determining such parameter. Based on experimentation, DBSCAN was able to correctly classify clusters and distinguish the clusters from noise. In comparison, CLARANS tended to divide big clusters, resulting to higher number of clusters.

The conclusion that DBSCAN performs better in cluster detection compared to other clustering algorithms confirms the results of the experiment done by Nerurkar, et al [18]. DBSCAN would thus be a good option for handling large data sets requiring clustering.

2.2 Related Theories

Clustering is an unsupervised machine learning method which groups a data set into disjointed clusters, the elements of which are of similar characteristics, and characteristically different from the elements of the other clusters [20]. This project utilized three clustering algorithms: (1) DBSCAN (Density-Based Spatial Clustering of Applications with Noise), (2) K-Means, and (3) Agglomerative hierarchical clustering.

2.2.1 DBSCAN

DBSCAN groups together closely packed data points and marks data points in low-density regions as outliers. It takes 2 hyperparameters: ***epsilon***, the maximum distance between 2 points for them to be considered closely packed and thus belong to the same cluster, and ***min_samples***, the minimum number of neighbors a point needs to have to be considered a core point (Figure 2.1).

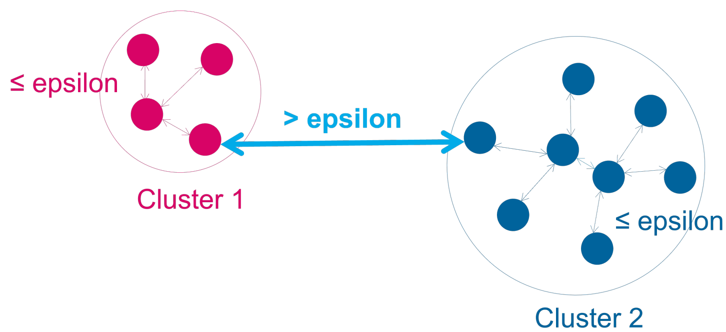

Figure 2.1: DBSCAN Hyperparameters

2.2.2 K-Means Clustering

K-means clustering takes the parameter ***k***, the number of clusters, as input. The number of clusters determines the number of centroids. The algorithm assigns each data point to the nearest centroid, forming the initial clusters. A new mean value of the centroid is then calculated, and the centroid is moved to this location. Data points are again assigned to the nearest centroid, and the mean value is recalculated. The process iterates until the centroid no longer moves (Figure 2.2).

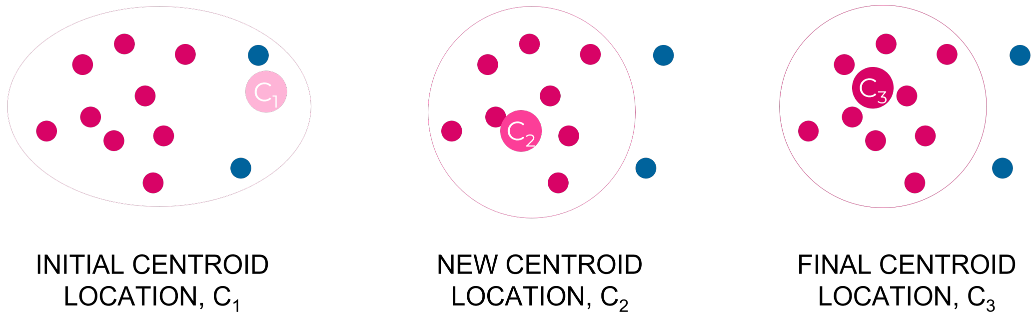

Figure 2.2: Changes in Centroid Location Using K-Means Clustering Algorithm

Compared to other algorithms, the time complexity of K-means is relatively low [21].

2.2.3 Agglomerative Hierarchical Clustering

In agglomerative hierarchical clustering, each data point is taken as a single-element cluster. The algorithm proceeds by combining two most similar clusters into a new bigger cluster. The iteration continues until all data points are grouped into a single big cluster (Figure 2.3).

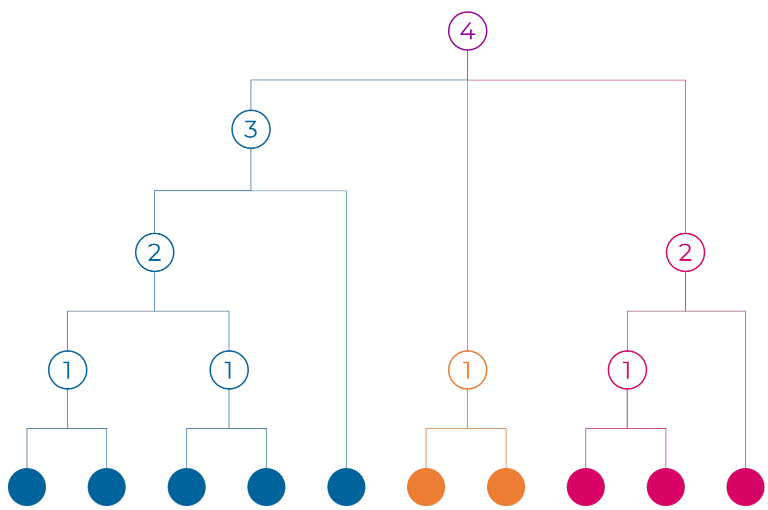

Figure 2.3: Iterative Combination of Similar Clusters in Agglomerative Clustering

2.3 Conceptual Framework

In this project, unsupervised machine learning algorithms are applied to Japanese population data taken during the pandemic, in order to generate coping profiles based on demographic, mental health, coping behavior, and COVID-19-related factors (Figure 2.4).

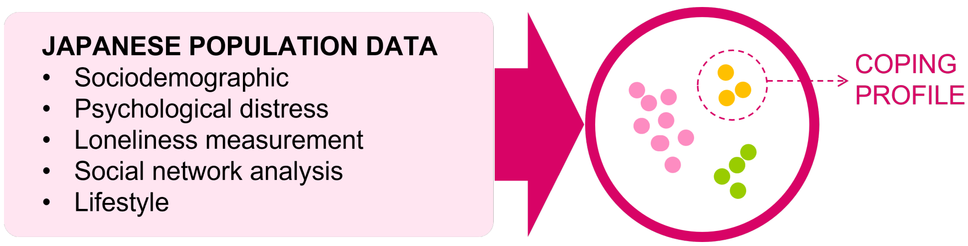

Figure 2.4: Conceptual Framework

Chapter 3

Methodology

The project involved four major steps: data collection, data preprocessing, model construction, and model interpretation (Figure 3.1).

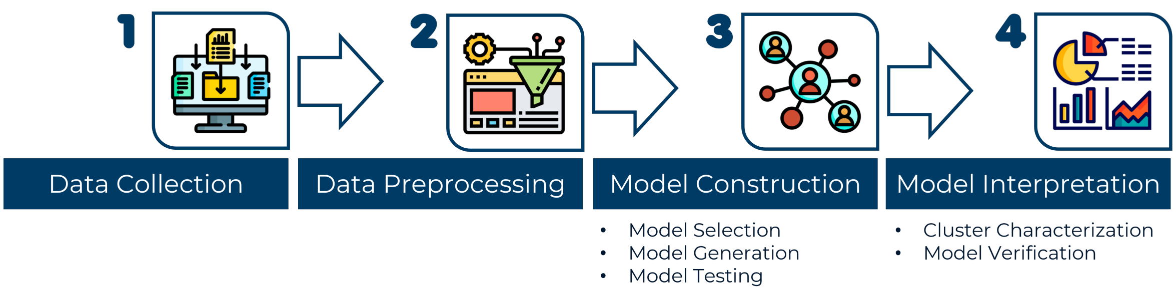

Figure 3.1: Methodology for Coping Profiles Clustering

3.1 Data Collection

Data was taken from an online survey conducted on the Japanese population from May 13 to 30, 2022, when the number of COVID-19-positive cases was high [22]. The dataset has 92 attributes, such as socio-demographic, COVID-19-related, mental health, and coping behavior information of 16,641 respondents (Refer to Tables 3.1-3.3 for the description of data).

Table 3.1: Socio-Demographic Data (Percentage Against Total of Respective Population) - Part 1/2

Parameter	**N(%)**		
	Total	**Male**	**Female**
Overall	16,641(100)	9,025(100)	7,616(100)
Age			
18-29	988(5.9)	217(2.4)	771(10.1)
30-49	6,763(40.6)	2,903(32.2)	3,860(50.7)
50-64	6,248(37.5)	3,987(44.2)	2,261(29.7)
≥ 65	2,642(15.9)	1,918(21.3)	724(9.5)
Occupation			
Employed	11,700(70.3)	7,075(78.4)	4,625(60.7)
Homemaker	2,366(14.2)	51(0.6)	2,315(30.4)
Student	120(0.7)	37(0.4)	83(1.1)
Unemployed	2,230(13.4)	1,723(19.1)	507(6.7)
Other	225(1.4)	139(1.5)	86(1.1)
Healthcare worker			
Self	980(5.9)	345(3.8)	635(8.3)
Family	1,101(6.6)	621(6.9)	480(6.3)
Marital status (Married)	10,762(64.7)	5,895(65.3)	4,867(63.9)
Annual household income (JPY)			
$<$ 2.0 million	1,323(8)	703(7.8)	620(8.1)
2.0–3.9 million	2,952(17.7)	1,596(17.7)	1,356(17.8)
4.0–5.9 million	3,223(19.4)	1,822(20.2)	1,401(18.4)
6.0–7.9 million	2,451(14.7)	1,417(15.7)	1,034(13.6)
$\geq$ 8.0 million	3,592(21.6)	2,207(24.5)	1,385(18.2)
Unknown	3,100(18.6)	1,280(14.2)	1,820(23.9)

Table 3.2: Socio-Demographic Data (Percentage Against Total of Respective Population) - Part 2/2

Parameter	**N(%)**		
	Total	**Male**	**Female**
Overall	16,641(100)	9,025(100)	7,616(100)
Treatment of severe physical diseases			
Current	818(4.9)	626(6.9)	192(2.5)
Previous	1,325(8)	920(10.2)	405(5.3)
Treatment of mental problems			
Current	918(5.5)	463(5.1)	455(6)
Previous	1,519(9.1)	742(8.2)	777(10.2)

The dataset was classified into four categories: demographics, COVID-19-related, coping mechanism, and mental health.

3.1.1 Demographic Data

Of the 16,641 respondents, 7,616 were females and 9,025 were males. Almost 80% were within the age range 30 to 64, and 70% were employed. The demographic data also included marital status, household income, and physical and mental illness history.

3.1.2 COVID-19-Related Data

The COVID-19-related data included interaction deterioration, economic deterioration, COVID-19 illness history, exposure to COVID-19 patients, and number of vaccinations. Interaction deterioration is a measure of how interpersonal relations of an individual worsened during the pandemic. Similarly, economic deterioration is a measure of how an individual's economic situation worsened in the same time period.

3.1.3 Coping Behavior Data

The coping behavior data are individual attitudes and ways of living to contend with and to overcome the difficulties of the pandemic. The data included optimism, trust in government, exercise, continuous prevention activities taken, and online and offline interactions.

3.1.4 Mental Health Data

The mental health data were taken using standard clinical instruments such as the UCLA Loneliness Scale (UCLA-LS3), Subjective Well-being (SHS), Kessler Psychological Distress Scale-6 (K6), and Post-Traumatic Growth Inventory (PTGI).

The **UCLA Loneliness Scale** assesses how often a person feels disconnected from other people. UCLA-LS3 utilizes a 10-item questionnaire, each item rated from 1 (never) to 4 (always). The total score ranges from 10 to 40, a higher score indicating higher degree of loneliness [23].

Subjective well-being was measured using the **Subjective Happiness Scale (SHS)**, an assessment of the overall happiness of an individual. SHS has 4 questions and uses a Likert-type rating scale from 1 (not at all) to 7 (completely). The score is calculated by dividing the total ratings by 4. Score ranges from 1 to 7, 7 indicating highest overall happiness [24].

Table 3.3: Data Related to Coping Behavior, Mental Health Status, and COVID-19 Impact

Data Feature	Description
Coping behavior	
Exercise	Amount of exercise
Continuous prevention	Level of practicing preventive measures against COVID-19 infection
Trust in government	Level of trust in the COVID-19 information provided by the government
Optimism	Level of positive thinking about the future
Mental health	
UCLA	Degree of loneliness measured using the University of California Los Angeles (UCLA) Loneliness Scale. It is based on the frequency of feeling of disconnection from others.
SHS	Subjective well-being. It is a measure of the overall happiness of an individual based on the Subjective Happiness Scale (SHS).
PTGI-X	Post-traumatic growth inventory. A measure of a person's growth and self-improvement after a trauma.
K6	Level of psychological stress measured using Kessler Psychological Distress Scale-6 (K6), based on a person's feelings and experiences for the past 30 days.
COVID-19-Related	
Interaction deterioration	How much one's interpersonal relations worsened during the pandemic
Economic deterioration	How much one's economic situation worsened during the pandemic

The **Kessler Psychological Distress Scale-6 (K6)** is a tool for quick assessment of serious mental illness, and measures psychological distress by evaluating one's feelings and experiences for the past 30 days. K6 utilizes a 6-item questionnaire, rated from 0 (none of the time) to 4 (all of the time). The total score ranges from 0 to 24. Individuals who score from 0 to 4 are classified as having no to low psychological distress, while those who score from 13 to 24 are classified as having serious psychological distress. Any score between these ranges is considered to be an indication of mild to moderate psychological distress [25].

To assess the growth and self-improvement a person experiences after a trauma or a stressful encounter, the **Post-Traumatic Growth Inventory (PTGI)** is used. PTGI considers 5 factors, namely personal strength, new possibilities, improved relationships, spiritual growth, and appreciation for life. The standard PTGI assessment is composed of 21 items, rated from 0 (I did not experience this as a result of my crisis) to 5 (I experienced this change to a very great degree as a result of my crisis). The score for each factor is calculated. The total PTGI score is the sum of the scores of all factors, and ranges from 0 to 105, with higher scores indicating better post-traumatic growth [26].

For this study, the PTGI data was taken using the **Post-Traumatic Growth Inventory-Expanded (PTGI-X)**, modified for the pandemic. It utilized 6 questions specifically designed for the COVID-19 pandemic experience, rated from 0 (Did not experience this change as a result of my crisis) to 6 (Experienced this change to a very great degree as a result of my crisis). The total score ranged from 0 to 36, the highest score indicating greatest post-traumatic growth. Each factor was also scored separately.

The survey also included mental health measurements for depression (PHQ-9), anxiety (GAD-7), Introsensory Perception (MAIA), alcohol dependence, and social network.

3.2 Data Preprocessing

Visualization of data was initially conducted to determine the required preprocessing steps. Preprocessing of the entire dataset involved exclusion of entries with unknown/blank personal or household income, and one-hot encoding of categorical variables. For binary categorical variables, only one of the data column was kept and utilized.

To select the appropriate data features, the dataset was further modified, resulting to 28 preprocessing versions, which were combinations of the following changes to the data:

- Inclusion/exclusion of some high-correlation features
- Feature standardization
- Dropping Prefecture, Student, and PTG subscales
- Scaling features to be within [0, 1]
- Using grouped Job variable

Grouped version of the variable job was used (job groupings: employed, homemaker, student, unemployed, others) because significant differences on social isolation and psychological distress are present among the job groups [27] [28].

3.3 Model Construction

Model construction required selection, generation, and testing of the clustering model.

3.3.1 Model Selection

DBSCAN was initially identified as the modelling algorithms because based on studies, it is better at cluster detection, has low sensitivity to noise, and is highly scalable [18] [19]. K-means and agglomerative clustering were also implemented as potential alternatives to DBSCAN due to the high number of outliers in the resulting DBSCAN model.

3.3.2 Model Generation

DBSCAN

The DBSCAN models were built using min_samples equal to multiples of the number of features in the preprocessed data. The range of epsilon values was dependent on the preprocessing version.

K-Means

The K-means models were built for k=2 to 10, using yellowbrick's KElbowVisualizer for the elbow method.

Agglomerative Clustering

The agglomerative models were built for k=2 to 10, using yellowbrick's KElbowVisualizer for the elbow method.

Elbow Method

To determine the optimal number of clusters, the elbow method was used. The elbow method fits the model with a range of values for the number of clusters [29]. The optimal number of clusters is the maximum beyond which there is no computational advantage, and there is no better modelling of the data. This is based on the distortion score, computed as the mean sum of squared distances from each point to its assigned centroid.

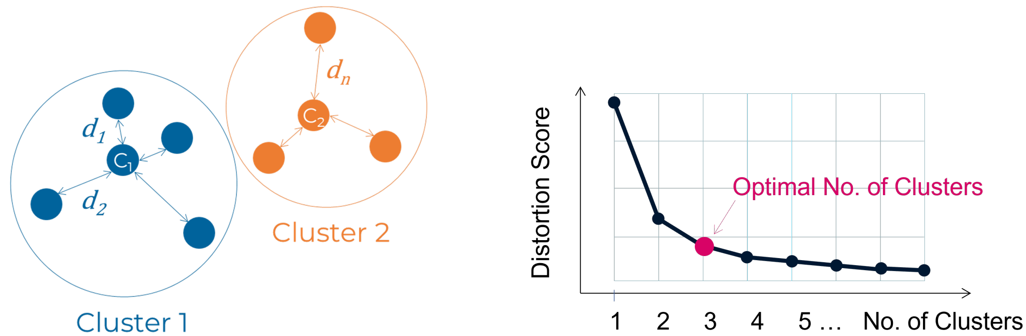

Figure 3.2: Conceptual Illustration of the Elbow Method

The distortion score is calculated as:

$$\text{distortion score} = \frac{d_1^2 + d_2^2 + ... + d_n^2}{n}$$

where

d_n = distance from point n to the centroid

n = number of points

3.3.3 Model Testing

Model performance was tested using silhouette score. Silhouette score is the mean of the silhouette scores of all data points in a dataset [30].

$$\text{silhouette score} = \frac{\text{sum of silhouette scores of all data points}}{\text{number of data points}}$$

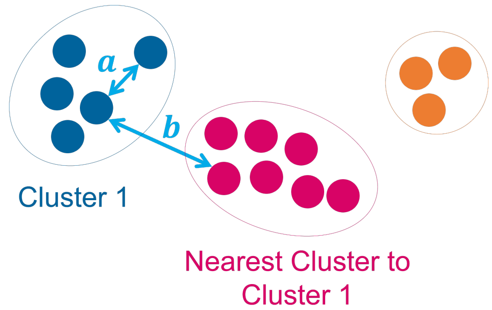

Figure 3.3: Reference Figure for Calculation of Silhouette Score

For each data point, the silhouette score is calculated as the average intra-cluster distance a and the average inter-cluster distance b:

$$\text{silhouette score} = \frac{b_{ave} - a_{ave}}{max(b_{ave}, a_{ave})}$$

where

a_{ave} = average distance of the data point to all other data points in the same cluster

b_{ave} = average distance of the data point to all data points in the nearest cluster

The silhouette score can range from -1 to 1, the higher score indicating higher cluster density and better separation of clusters. A silhouette score of -1 indicates the possibility of incorrect cluster assignment.

3.4 Model Interpretation

After modelling, the clusters were characterized by

- Plotting the standardized mean feature values on a radar plot, and
- Viewing the feature value distribution through violin plots.

The resulting clusters were validated by psychology experts from Japan:

- Dr. Tetsuya Yamamoto of Clinical Psychoinformatics Laboratory, Tokushima University
- Dr. Chigusa Uchiumi of Graduate School of Technology, Industrial and Social Sciences, Tokushima University
- Dr. Nagisa Sugaya of Department of Public Health, School of Medicine, Yokohama City University

Chapter 4

Coping Profile Clustering

Building viable clustering models requires a combination of the appropriate preprocessed data and model hyperparameters. These, utilized as input to the elbow method, determine the optimal number of clusters. The silhouette score then serves as a metric for testing the consistency of clustering.

4.1 Model Building

Each data preprocessing version was used in combination with different values of model parameters, in order to come up with the best silhouette score. The elbow method was further applied on the agglomerative and K-means models to determine the optimal number of clusters (Figure 4.1).

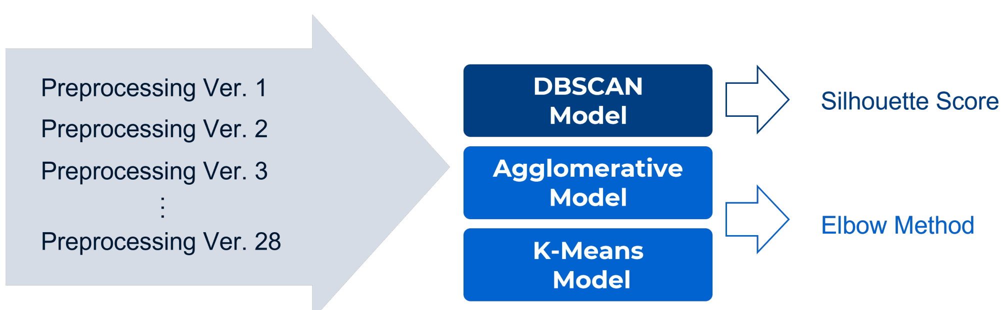

Figure 4.1: Experimentation with Data Preprocessing and Model Parameters

Table 4.1 shows the description of each data preprocessing version, and the data preprocessing involved. Cell shading indicate the change implemented on the data.

4.2 Results

4.2.1 DBSCAN Modelling Results

The resulting best silhouette scores and the corresponding number of clusters from the DBSCAN experimentations are shown in Figure 4.2.

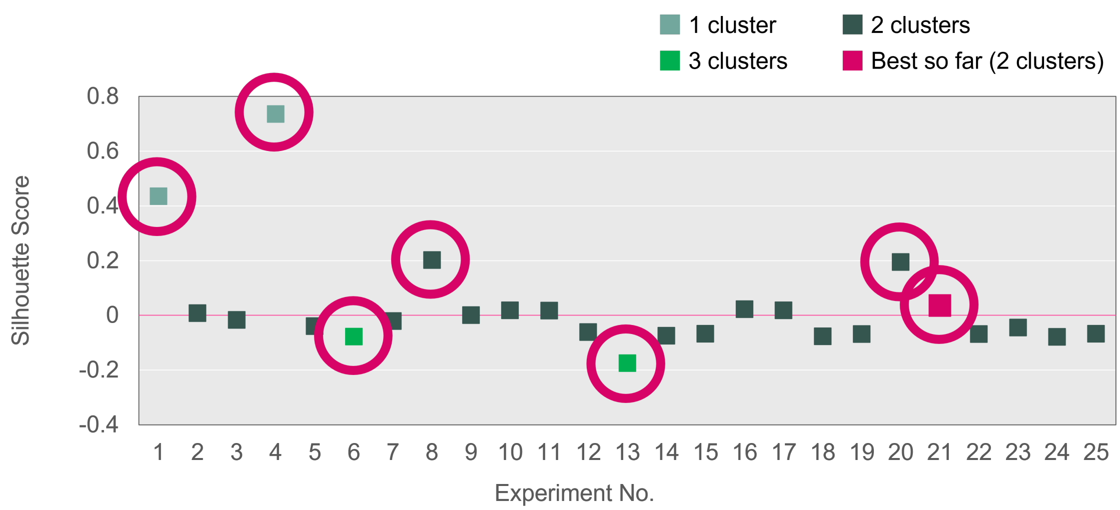

Figure 4.2: DBSCAN Modelling Experiment Results

Experiments 1 and 4 have high silhouette scores, but only 1 cluster. While Experiments 6 and 13 generated 3 clusters, their silhouette scores are negative. Experiments 8 and 20 have 2 clusters, but are dependent on one variable only. Among the other experiments with 2 clusters, Experiment 21 has the highest silhouette score.

Table 4.1: Data Preprocessing Versions

Ver	Description	Drop high correlation	Standardized	Drop prefecture	MinMax [0, 1]	Drop student	Drop PTG subscales	Use grouped variables	Drop yeays enrolled
1	One-hot encoding, feature selection (but high correlation variables kept), & exclusion of entries with unknown/blank income								
2	One-hot encoding, feature selection (but some high correlation variables dropped), & exclusion of entries with unknown/blank income								
3	Version 1, but with standardization (all features)								
4	Version 2, but with standardization (all features)								
5	Version 1, but drop prefecture features								
6	Version 2, but drop prefecture features								
7	Version 3, but drop prefecture features								
8	Version 4, but drop prefecture features								
9	Version 1, but only numerical features are standardized		Numerical						
10	Version 2, but only numerical features are standardized		Numerical						
11	Version 9, but drop prefecture features		Numerical						
12	Version 10, but drop prefecture features		Numerical						
13	Version 8, but minmax scaling instead of standardization								
14	Version 10, but drop student & PTG subscales		Numerical						
15	Version 11, but drop student & PTG subscales		Numerical						
16	Version 10, but drop student		Numerical						
17	Version 11, but drop student		Numerical						
18	Version 10, but drop PTG subscales		Numerical						
19	Version 11, but drop PTG subscales		Numerical						
20	Version 8, but use grouped job instead							5_JOB_group	
21	Version 10, but use grouped job instead		Numerical					5_JOB_group	
22	Version 11, but use grouped job instead		Numerical					5_JOB_group	
23	Version 12, but use grouped job instead		Numerical					5_JOB_group	
24	Version 21, but drop student		Numerical					5_JOB_group	
25	Version 22, but drop student		Numerical					5_JOB_group	
26	Version 23, but drop student & PTG subscales		Numerical					5_JOB_group	
27	Version 14, but drop years enrolled		Numerical						
28	Version 22, but drop years enrolled		Numerical					5_JOB_group	

Figure 4.3 shows the characteristics of the clusters generated from DBSCAN Experiment 21. Clustering consists of 2 clusters, with 7,505 outliers represented by Cluster 1. Cluster 2 (n=439) has lower PTG (Posttraumatic Growth) scores and lower coping strategy scores compared to Cluster 0 (n=5471).

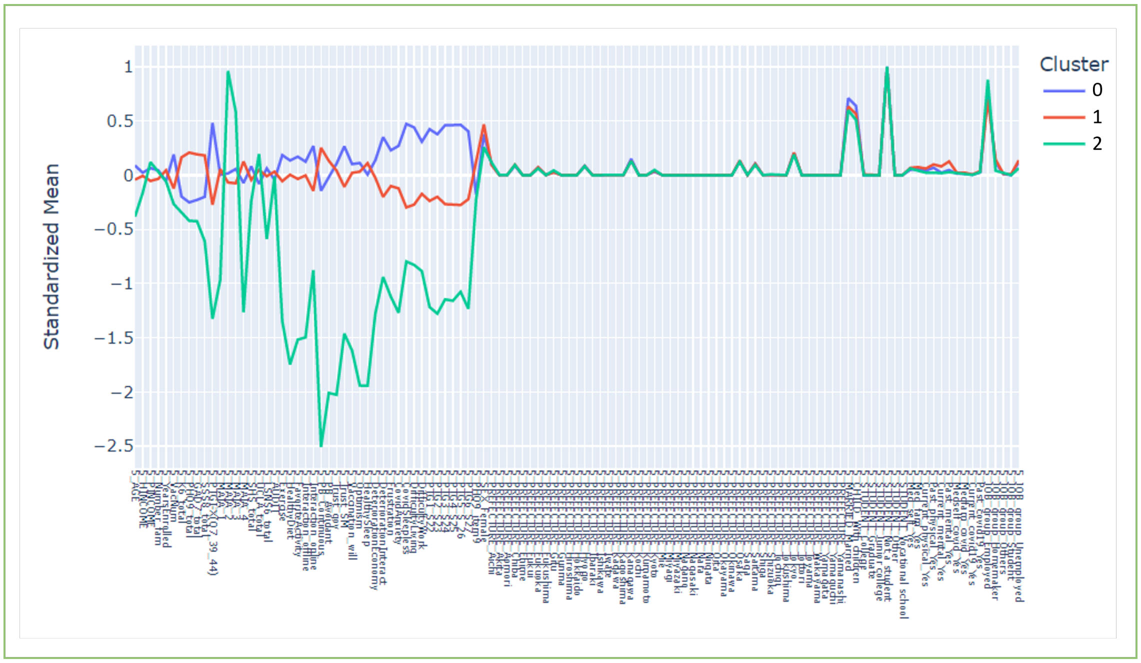

Figure 4.3: DBSCAN Experiment 21 Cluster Characteristics

4.2.2 K-Means Modelling Results

The best silhouette scores and the corresponding number of clusters generated by the K-means experimentations are shown in Figure 4.4.

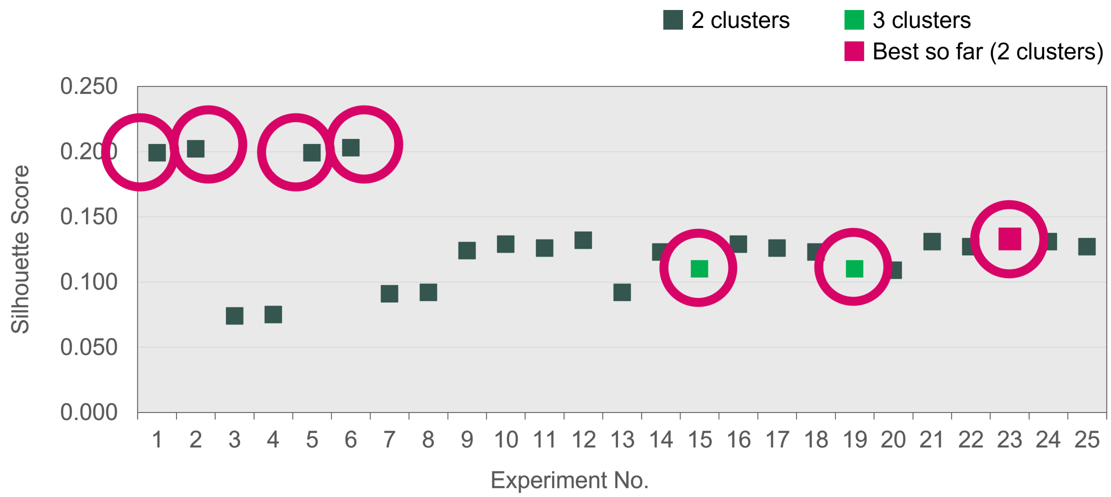

Figure 4.4: K-Means Modelling Experiment Results

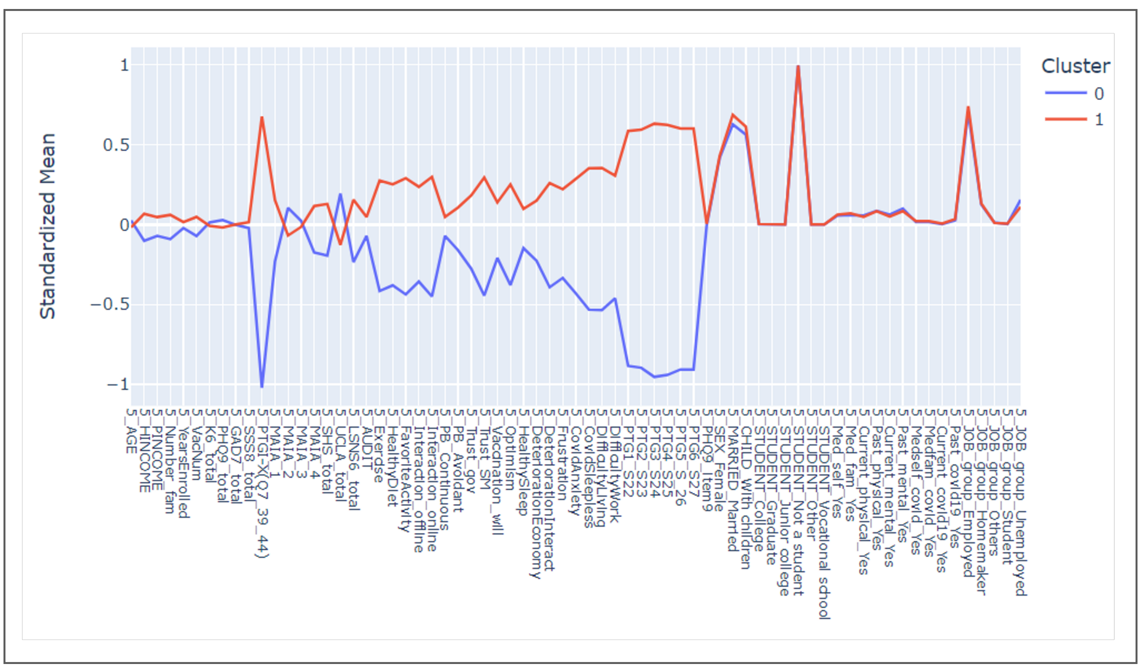

Figure 4.5: K-Means Experiment 23 Cluster Characteristics

In Figure 4.4, Experiments 1, 2, 5, and 6, have high silhouette scores, but are dependent on the variable age only. Experiment 23 has the highest silhouette score among the experiments with 2 clusters. Although Experiments 15 and 19 have 3 clusters, they have lower silhouette scores compared to Experiment 23.

As shown in Figure 4.5, K-Means Clustering 23 consists of 2 clusters. Cluster 0 (n=5342) has lower PTG (Posttraumatic Growth) scores and lower coping strategy scores compared to Cluster 1 (n=8073).

Clusters from Application of Elbow Method on K-Means Model

Application of elbow method on the same dataset determined 5 as the optimal number of clusters. Data Preprocessing version 22 has the highest silhouette score at 0.095, with a distortion score of 523857.754 (Figure 4.6).

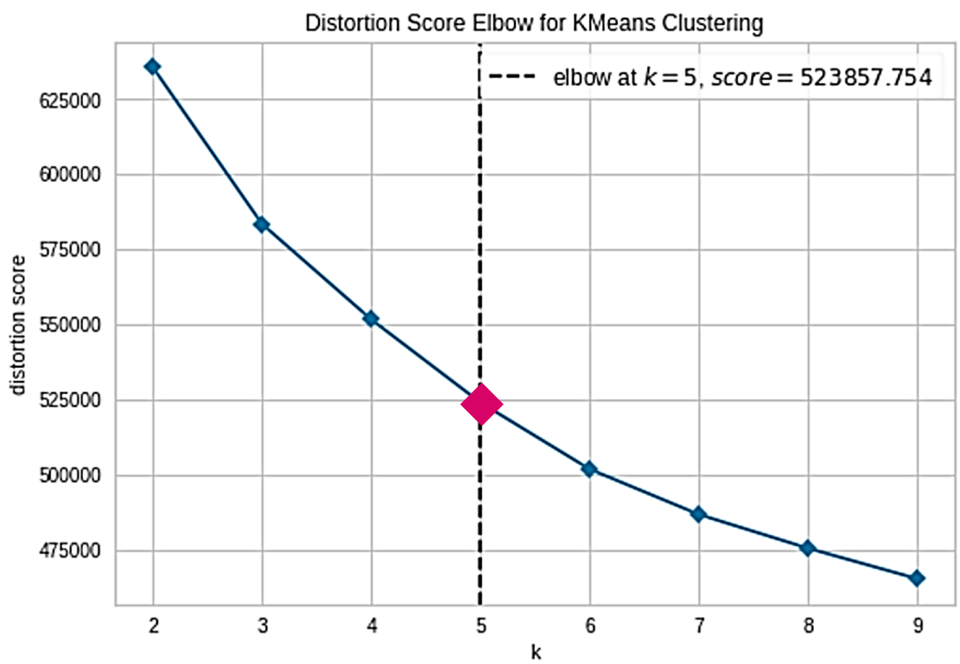

Figure 4.6: Result of Elbow Method Application on K-Means Model, Data Preprocessing Version 22, Elbow at k=5

4.2.3 Confirmation of K-Means Clusters using DBSCAN Modelling

DBSCAN modelling was also applied on data Preprocessing version 22 to find parameters that could show similar clusters as K-means. As can be seen in Figure 4.7, DBSCAN modelling resulted to 2 clusters, with 6,770 outliers (represented by Cluster -1 in the graph). Cluster 1 (n=400) has lower PTG scores, lower economic and interaction deterioration, and lower coping strategy scores compared to Cluster 0 (n=6,245).

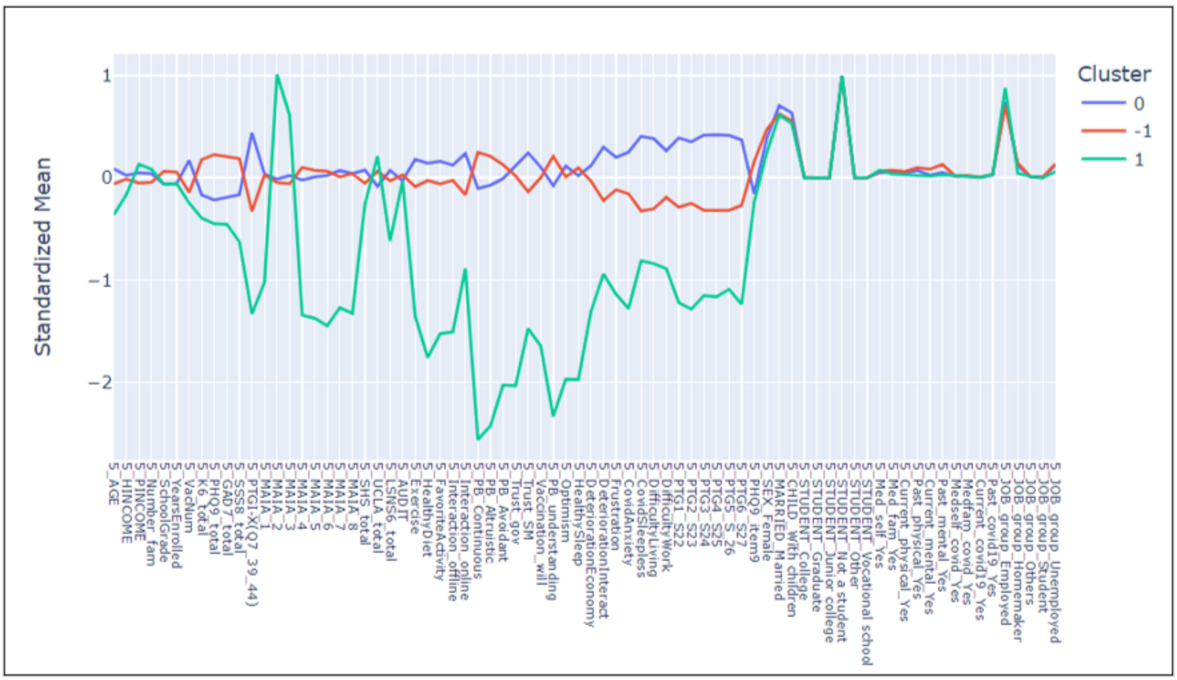

Figure 4.7: DBSCAN Modelling Clusters Using Data Preprocessing Version 22

The DBSCAN model has a silhouette score of 0.052, and has significantly higher number of outliers compared to the number of members in Cluster 1.

4.2.4 Confirmation of K-Means Clusters using Agglomerative Modelling

Agglomerative clustering was further applied on the dataset to attempt to generate clusters similar to the output of K-Means modelling. Using Elbow Method, data Preprocessing Version 14 resulted to the best silhouette score, with 2 clusters (Figure 4.8). This model has a silhouette score of 0.155, and a Davies-Bouldin Index of 2.343.

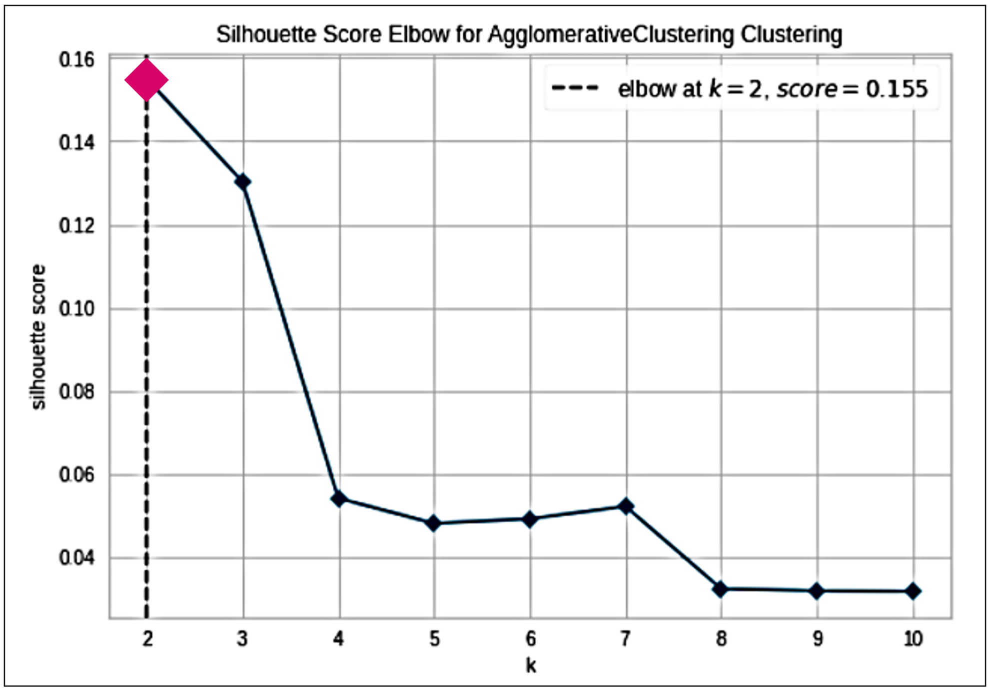

Figure 4.8: Result of Elbow Method Application on Agglomerative Model, Data Preprocessing Version 14, Elbow at k=2

Figure 4.9 shows the cluster characteristics of the Agglomerative model. Cluster 1 (n=2,194) has worse mental health scores, higher economic and interaction deterioration, and lower coping strategy scores compared to Cluster 0 (n=11,221).

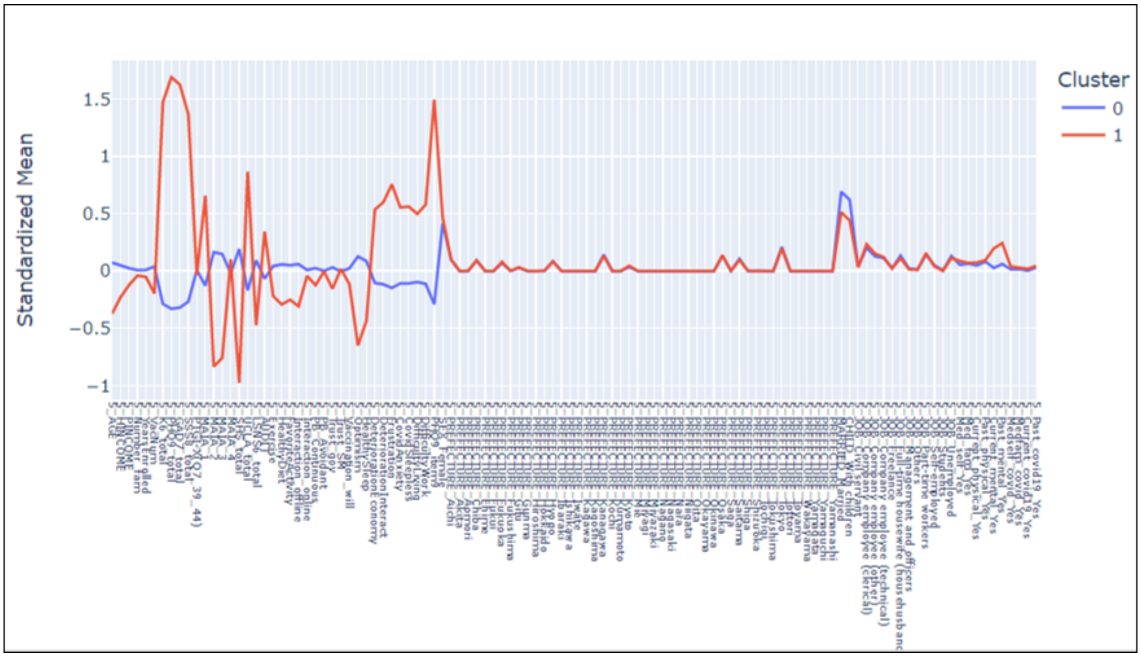

Figure 4.9: Agglomerative Modelling Clusters Using Data Preprocessing Version 14

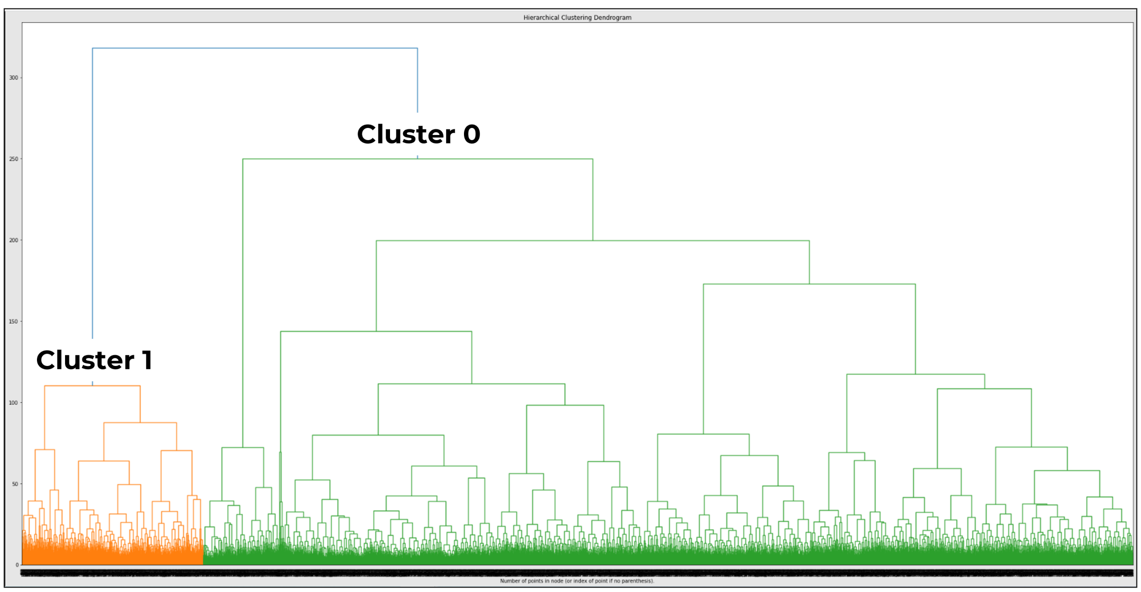

Figure 4.10: Dendrogram of the Agglomerative Model, Data Processing Version 14

The dendrogram of the agglomerative model (Figure 4.10) shows the merging of data points into clusters. Based on the dendrogram, Cluster 0 is more spread out than Cluster 1.

4.2.5 Characterization of the 5-Cluster K-Means Model

Both the DBSCAN and agglomerative models had 2 clusters. Using DBSCAN, almost half of the data points were classified as outliers. Of the three modelling algorithms used, only K-means resulted to 5 clusters. In terms of segmenting the population for the purpose of behavioral intervention, 5 clusters seemed more viable and more logical, so the 5-cluster K-means model was chosen for further characterization.

Radar Plot of Clusters

Figure 4.11 shows the radar plot of the distinguishing data features of the 5 clusters from K-Means modelling. The data features are color-coded, with coping behavior features colored purple, mental health blue, and COVID-19-related yellow. Data features with an asterisk are negative variables, the higher values indicating worse condition.

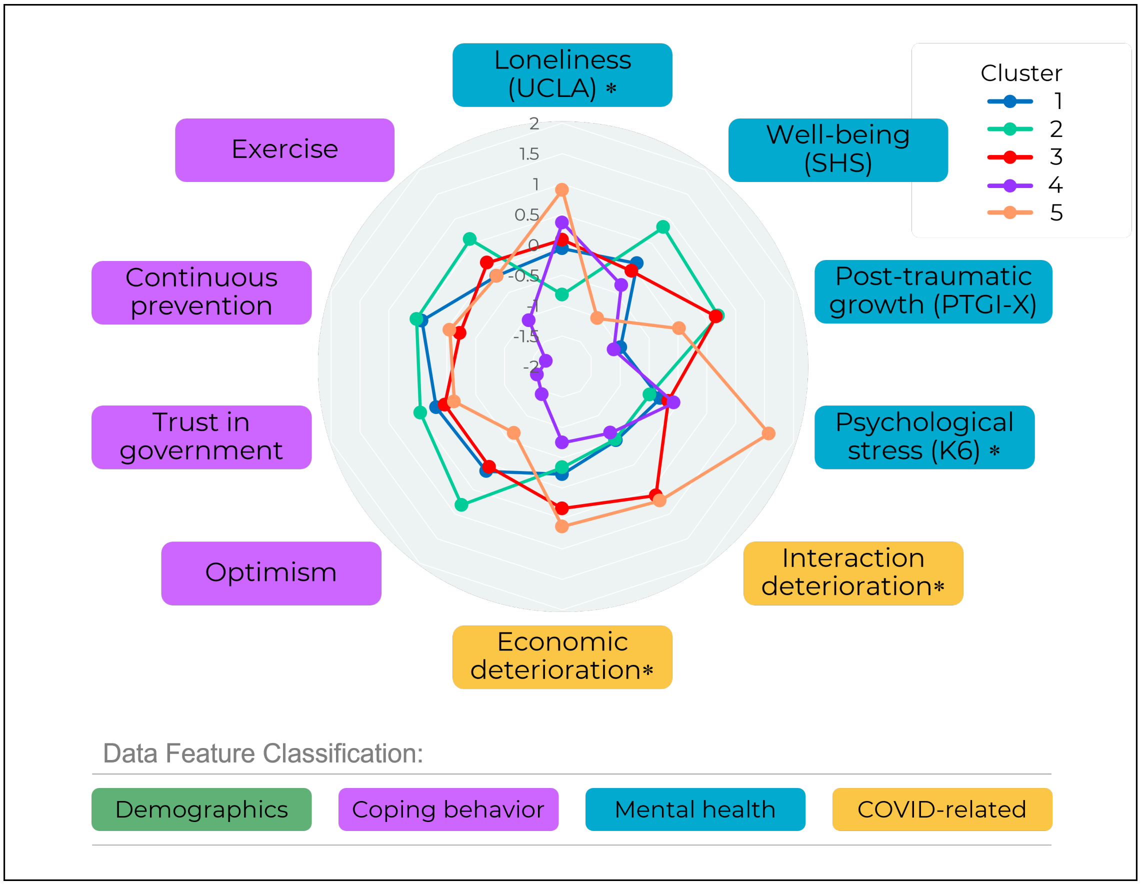

Figure 4.11: Radar Plot of the 5 Clusters from K-Means Modelling

Ten data features define the clusters:

- Coping behavior: Exercise, continuous prevention, trust in government, and optimism
- Mental health: Loneliness, well-being, Post-Traumatic Growth (PTG), and psychological stress
- COVID-19-related: Interaction deterioration and economic deterioration

No demographic features characterize the clusters.

From the radar plot, it can be observed that:

(1) Cluster 1 has low posttraumatic growth, but high continuous prevention and trust in government.

(2) Cluster 2 has better mental health scores, better coping scores, and less economic and interaction deterioration.

(3) Cluster 3 has more economic and interaction deterioration, but good posttraumatic growth.

(4) Cluster 4 has less economic and interaction deterioration, but low coping scores and posttraumatic growth.

(5) Cluster 5 has worse mental health scores, worse coping scores, and more economic and interaction deterioration.

Violin Plots of Data Features

Violin Plots of Coping Behavior Data Features of the 5 Clusters. The violin plots of the coping behavior data of the 5 clusters are shown in Figure 4.12. Compared to the other clusters, the coping behavior scores of Cluster 4 are more concentrated on the lower end of the violin plots.

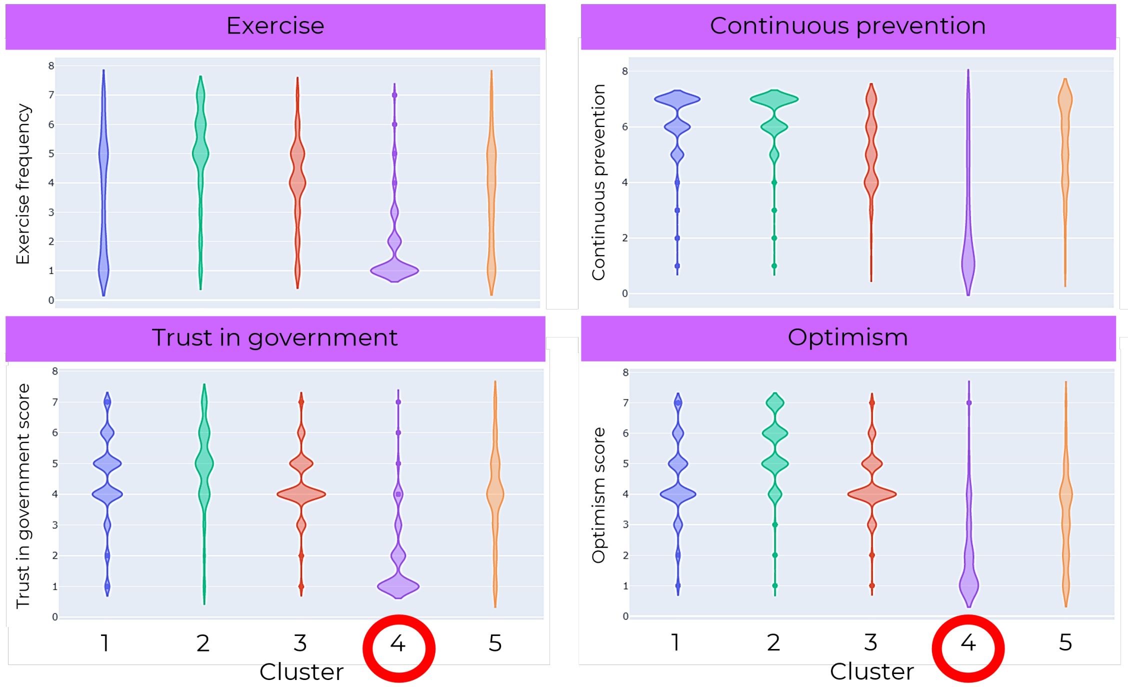

Figure 4.12: Violin Plots of Coping Behavior Data Features of the 5 Clusters

Violin Plots of Mental Health Data Features of the 5 Clusters. In Figure 4.13, it can be seen that the posttraumatic growth and psychological stress scores of Cluster 5 are more spread out compared to the other clusters.

Violin Plots of COVID-19-Related Data Features of the 5 Clusters Figure 4.14 shows the violin plots of the COVID-19-related data features of the 5 clusters. For Clusters 1, 2, and 4, interaction deterioration scores are more concentrated at the lower end of the violin plots. For Clusters 3 and 5, both the interaction deterioration and economic deterioration scores are concentrated at the middle, with Cluster 5 data more spread out.

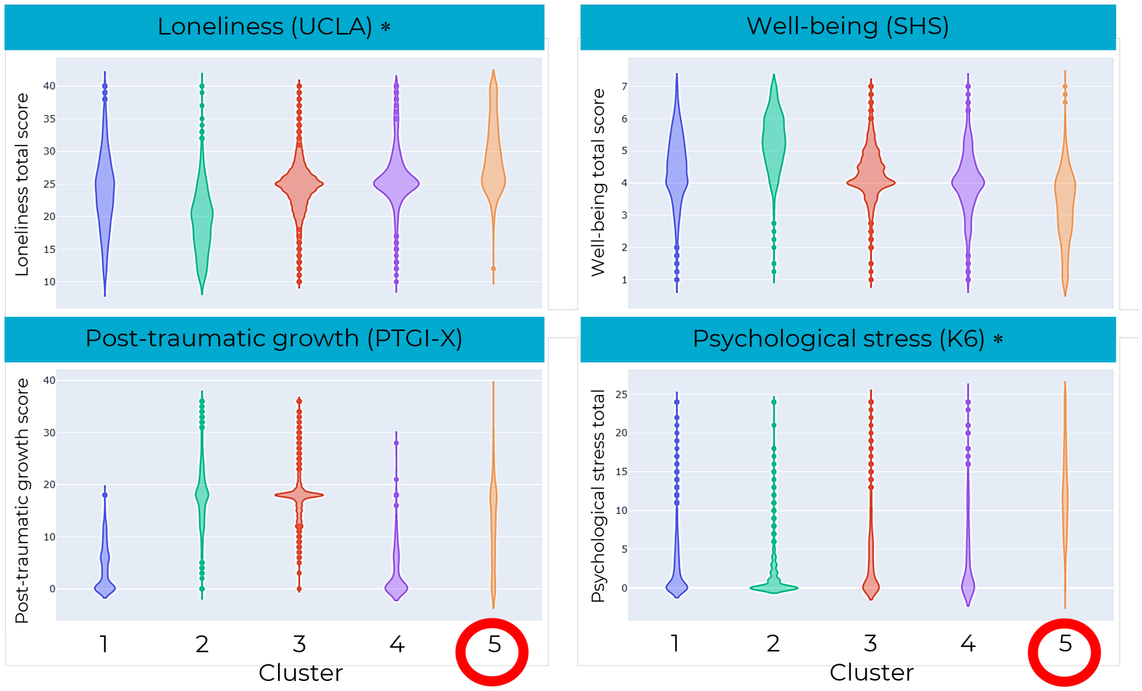

Figure 4.13: Violin Plots of Mental Health Data Features of the 5 Clusters

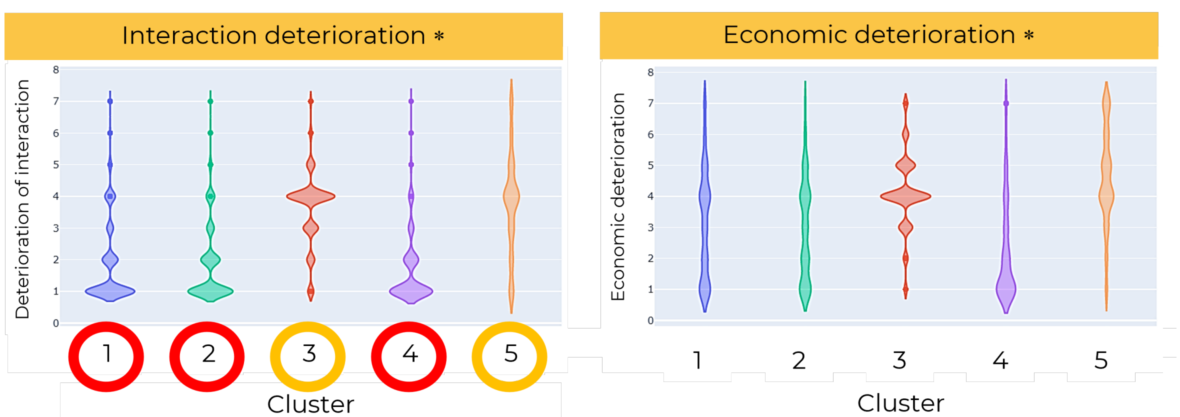

Figure 4.14: Violin Plots of COVID-Related Data Features of the 5 Clusters

4.2.6 Validation of the 5-Cluster K-Means Model

The resulting 5 clusters of the K-Means model were validated and accepted as meaningful by psychology experts from Japan, Dr. Tetsuya Yamamoto of Clinical Psychoinformatics Laboratory, Tokushima University, Dr. Chigusa Uchiumi of Graduate School of Technology, Industrial and Social Sciences, Tokushima University, and Dr. Nagisa Sugaya, Department of Public Health, School of Medicine, Yokohama City University.

4.2.7 Limitations of the Study

The validated 5-cluster K-Means model has relatively low silhouette score (0.095). Succeeding experimentation was not able to successfully increase the value of the silhouette score. As such, further study should be conducted to computationally validate the model.

Chapter 5

Discussion

To interpret the model, the cluster characteristics were compared with results of previous studies, thereby further defining the characteristics of the 5 clusters.

5.1 Optimal Number of Clusters

From the K-means modelling, the resulting highest silhouette score was 0.095, with 5 clusters as optimum. Cluster composition are as follows: Cluster 1: 3,403, Cluster 2: 2,883, Cluster 3: 3,952, Cluster 4: 1,149, and Cluster 5: 2,028.

Ten features were found to define the clusters:

• Coping behavior-related: Exercise, continuous prevention, trust in government, optimism

• Mental health-related: Loneliness (UCLA), well-being (SHS), post-traumatic growth (PTGI), psychological stress (K6)

• COVID-19 pandemic-related: Interaction deterioration, economic deterioration.

No demographic parameter distinctly defined the clusters. The interaction of demographic features with the other features defining the clusters needs further study.

5.2 Cluster Characteristics

The radar plots of the resulting clusters from K-means modelling are shown in Figures 5.1 to 5.5. The features marked with an asterisk (loneliness, psychological stress, interaction deterioration, and economic deterioration) are negative variables, with higher values indicating negative psychological characteristics.

5.2.1 Cluster 2: Optimistic

Cluster 2 (Figure 5.1) has the highest optimism score, trusts the government the most, and has high continuous prevention scores. Likewise, it has the highest PTG scores. Expectedly, Cluster 2 has the highest well-being and the lowest loneliness scores.

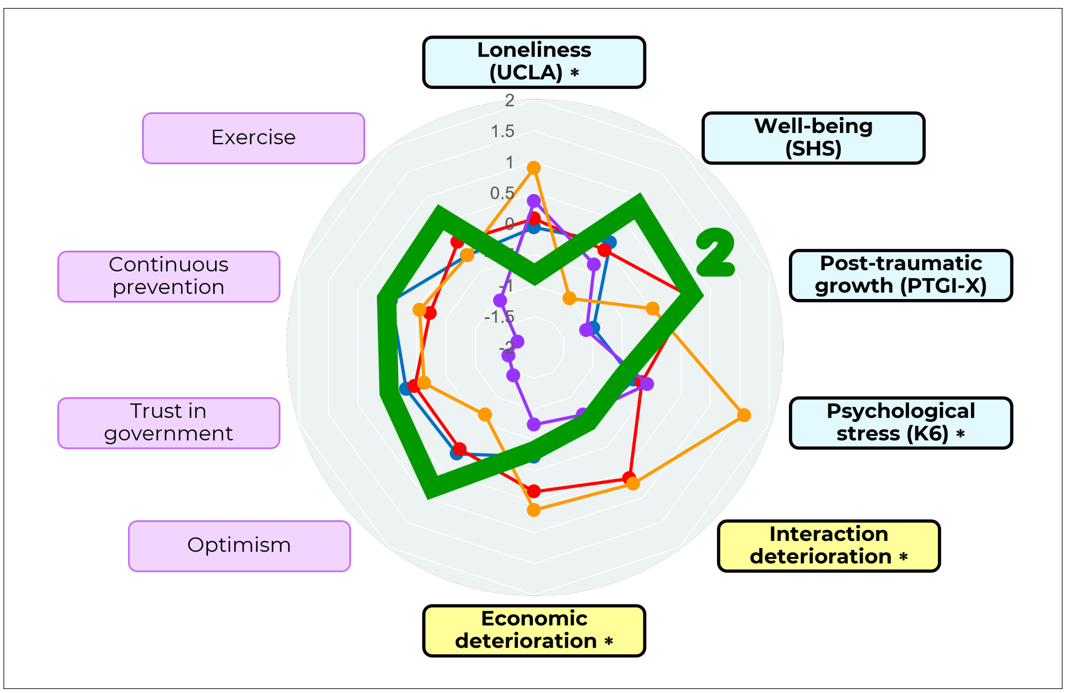

Figure 5.1: Radar Plot of Cluster 2. Cluster 2 has high mental health and coping scores, and less economic and interaction deterioration.

Studies suggest that the pandemic has a positive psychological effect on some people [31], and that optimists are more likely to benefit from a trauma [32], which may explain the high PTG scores of Cluster 2. Moreover, among the clusters, Cluster 2 has the highest exercise scores. A survey of the Spanish population indicated that higher PTG scores are positively correlated with physical exercise and number of leisurely hours spent by individuals [33]. It can thus be inferred that Cluster 2 is composed of people who gained positive psychological benefit from the pandemic.

5.2.2 Cluster 4: Pessimistic

Cluster 4 is characterized by pessimism (Figure 5.2). Like Cluster 2, Cluster 4 has less economic and interaction deterioration, but has the lowest optimism score, lowest trust in government, and lowest continuous prevention. PTG is also the lowest. People belonging to Cluster 4 are less likely to positively derive from the pandemic because of their overall negative outlook. This corresponds with a study by Chang, Maydeu-Olivares and D'Zurilla [34] which found direct correlation between pessimism and depressive symptoms among college students. Moreover, it was observed that optimists exhibit less distress and cope actively under trauma [35].

5.2.3 Cluster 1: Semi-optimistic/semi-pessimistic

The characteristics of Cluster 1 (Figure 5.3) are between that of Clusters 2 and 4. Both Clusters 1 and 2 encountered similar economic and interaction deterioration, but Cluster 2 has significantly lower loneliness and higher well-being scores. That Cluster 1 has low PTG scores (second lowest to Cluster 4) may be the cause of its low well-being scores.

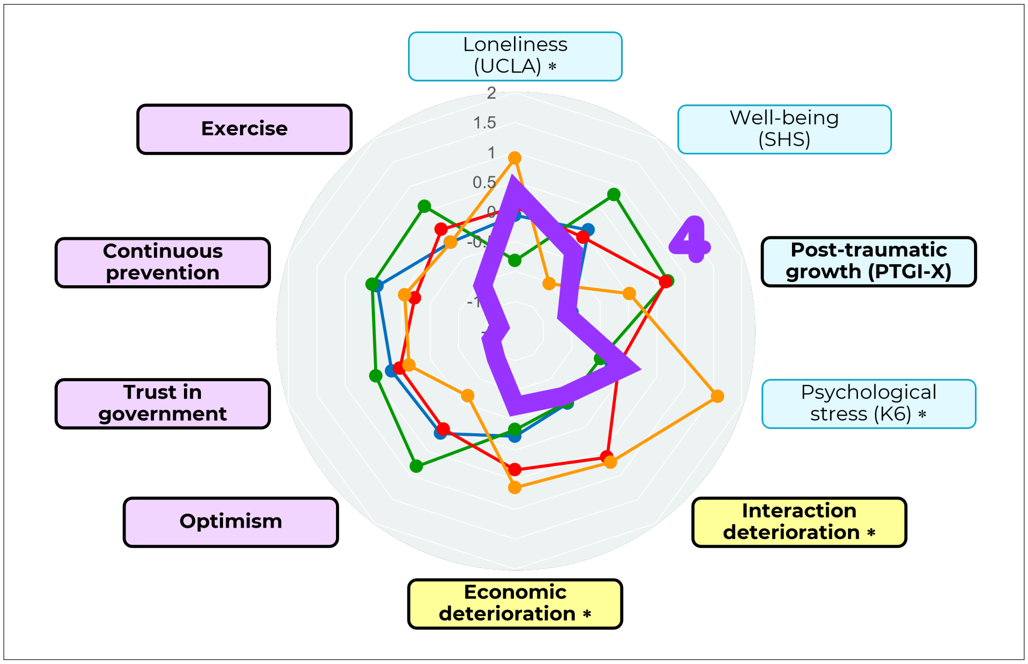

Figure 5.2: Radar Plot of Cluster 4. Cluster 4 has low economic and interaction deterioration, but low coping scores and post-traumatic growth.

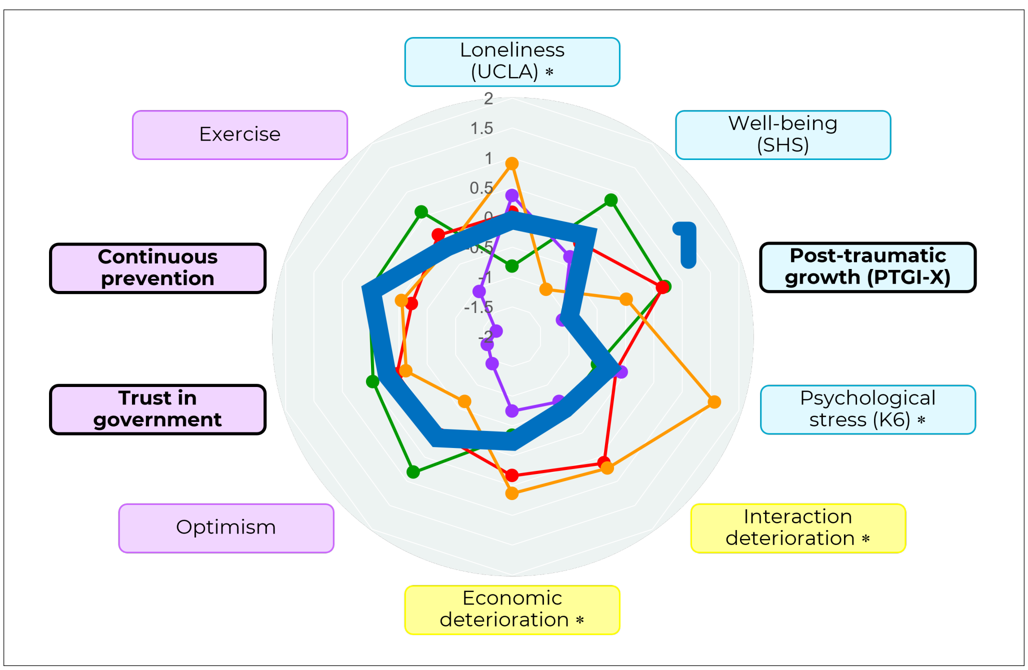

Figure 5.3: Radar Plot of Cluster 1. Cluster 1 has low post-traumatic growth, but high continuous prevention and trust in government scores.

5.2.4 Cluster 3: Resilient

Cluster 3 (Figure 5.4) has high PTG scores, and perceived economic and interaction deterioration.

Although Cluster 2 and Cluster 3 have the topmost PTG scores, Cluster 2 has comparatively higher well-being and lower stress scores. These differences can be attributed to the economic and interaction deterioration experienced by Cluster 3. However, Blom, et al. [36] reported that there are actually two subgroups of people who experience post-traumatic growth, those who cope adaptively, and those who cope by rumination. People who cope adaptively perceive the pandemic and the resulting stress to have minimal effect on life. Contrarily, people who cope by rumination may feel hugely affected by stress and the pandemic, and still experience PTG. Cluster 2 can be considered to belong to the group that cope adaptively, and Cluster 3, to the group that cope by rumination. This difference in perception can account for the gap in the well-being scores of the two clusters, despite similarly high PTG scores. Because individuals in Cluster 3 experienced PTG, they are able to cope with the pandemic and are considered resilient.

5.2.5 Cluster 5: Highly-Stressed

Cluster 5 (Figure 5.5) has the highest loneliness, highest psychological stress, and lowest well-being scores.

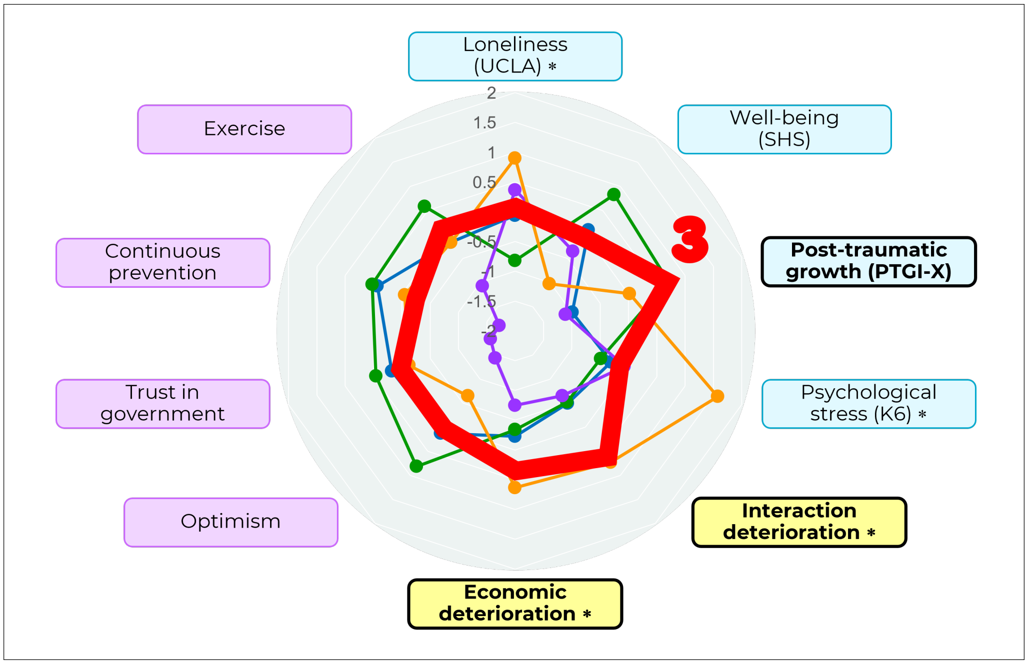

Figure 5.4: Radar Plot of Cluster 3. Cluster 3 has high economic and interaction deterioration, but good post-traumatic growth.

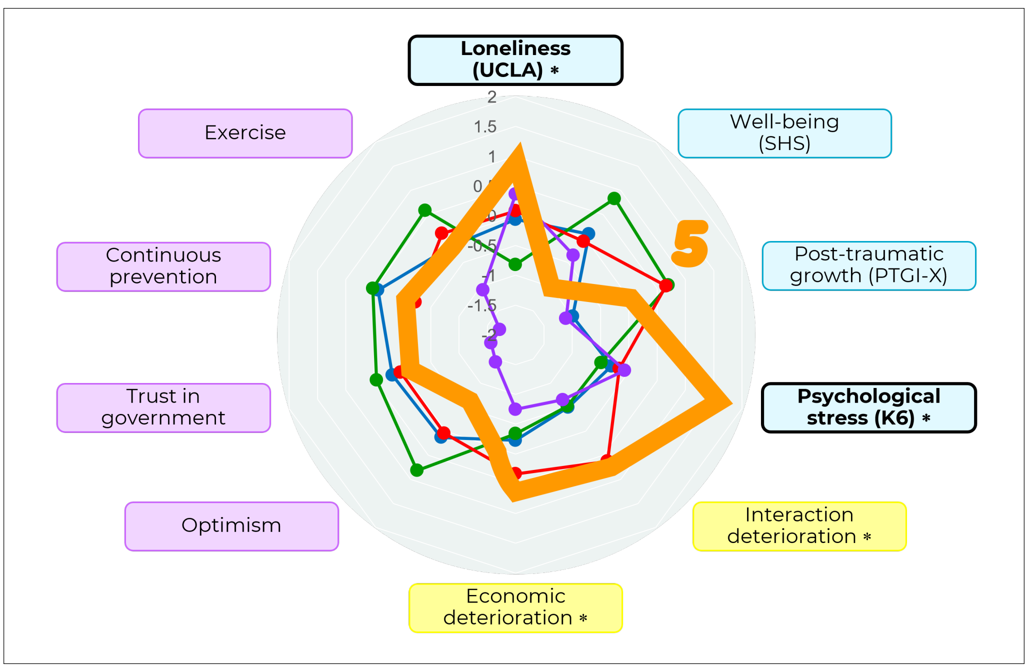

Figure 5.5: Radar Plot of Cluster 5. Cluster 5 has low mental health scores and coping scores, and high economic and interaction deterioration.

Both Cluster 3 and Cluster 5 have high deterioration scores, but Cluster 5 scores high on loneliness. Unlike Cluster 3, Cluster 5 has low PTG scores. A meta-analysis conducted by Helgeson, Reynolds and Tomich [32] indicated that higher PTG scores are associated with positive well-being and less depression, possibly accounting for the loneliness of Cluster 5. This cluster represents those who found difficulty adapting to and recovering from the stress induced by the pandemic.

5.3 Comparison of the Clustering Models

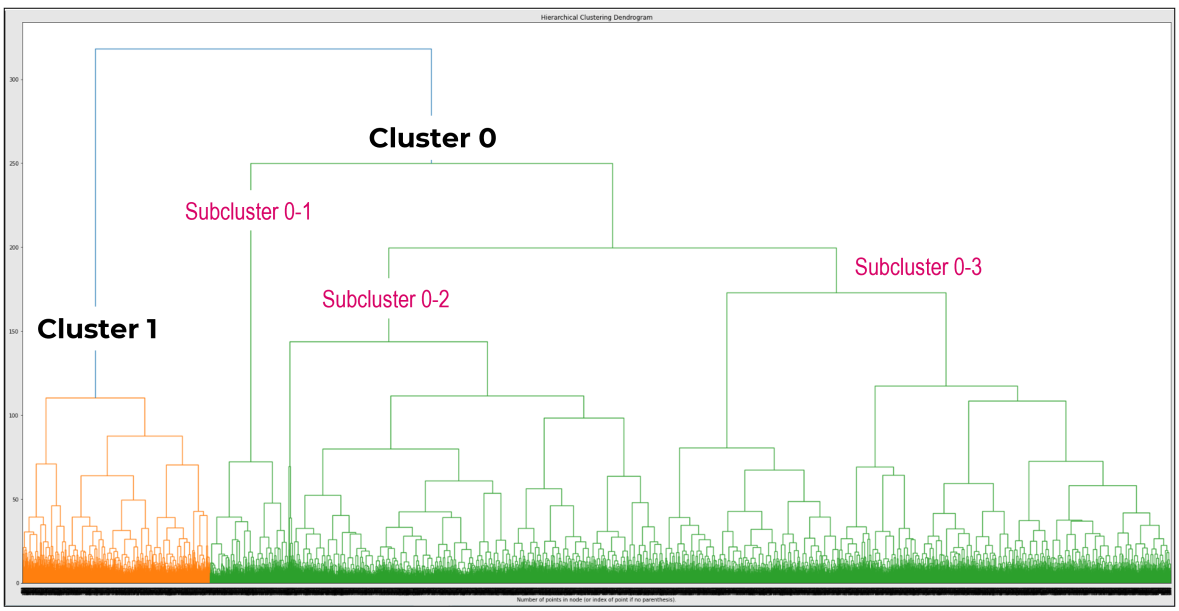

Figure 5.6: Dendrogram of the Agglomerative Model Showing Subclusters 0-1, 0-2, and 0-3

The dendrogram of the agglomerative model shows that the data of Cluster 0 is quite spread apart (see Figure 5.6, Subclusters 0-1, 0-2, and 0-3) compared to Cluster 1, an indication of the difference in their densities. Since DBSCAN creates clusters based on the parameter *eps*, which defines the maximum point-to-point distance within a cluster, it can be deduced that DBSCAN clustering can potentially be impacted by this difference in cluster densities. The low density of Cluster 0 may also be the cause of the high number of outliers of the DBSCAN model. Contrarily, it is possible that since K-means forces groupings based on the number of centroids regardless of density, K-means was able to generate better clusters from the dataset.

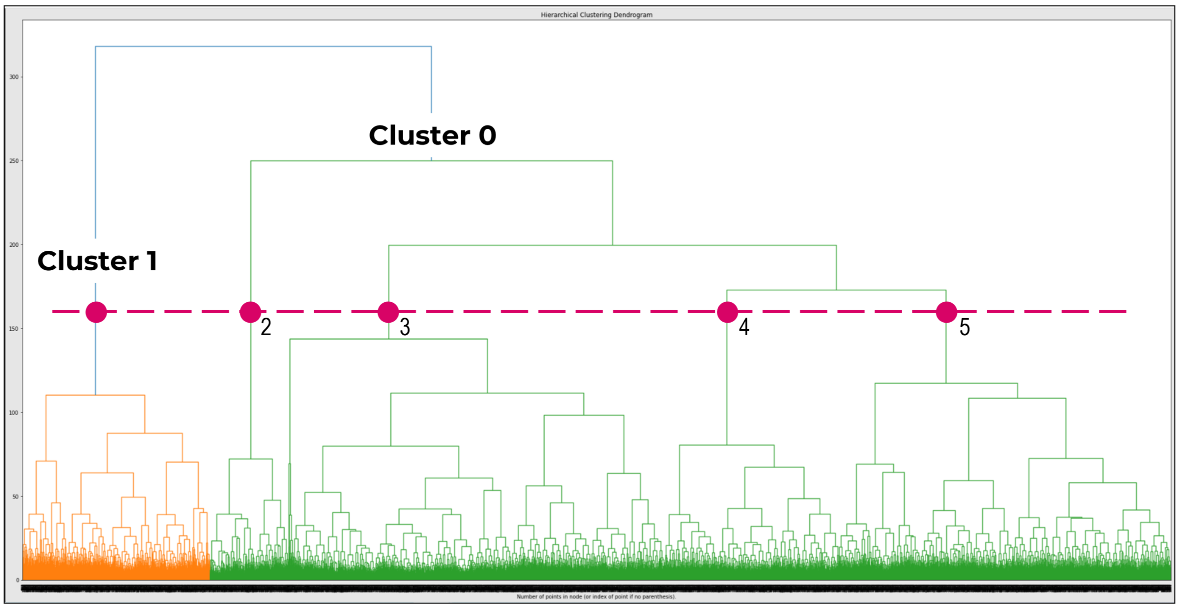

Figure 5.7: Dendrogram of the Agglomerative Model Showing 5 Clusters at Lower Distance Threshhold

The 2-cluster agglomerative model was the result of applying elbow method on the dataset. The criteria for agglomerative clustering is the number of clusters or the distance threshold (distance "at or above which clusters will not be merged" [37]). Using the Elbow method, the criteria used was the number of clusters and silhouette score. From the dendrogram in Figure 5.7, it can be inferred that when a lower distance threshold is used, it would be possible for agglomerative clustering to generate 5 clusters, the same number of clusters generated through K-Means. As such, agglomerative model using distance threshold as criteria may further be explored.

Chapter 6

Summary and Conclusion

Using the K-means clustering algorithm and the elbow method, there were 5 identified clusters of the population, characterized by 10 data features, namely exercise, continuous prevention, trust in government, optimism, loneliness (UCLA), well-being (SHS), post-traumatic growth (PTGI), psychological stress (K6), interaction deterioration, and economic deterioration.

The clusters can be generally described as semi-optimistic/semi-pessimistic (Cluster 1), optimistic (Cluster 2), resilient (Cluster 3), pessimistic (Cluster 4), and highly-stressed (Cluster 5). Post-traumatic growth was observed to be a determining factor that drives the characteristics of each cluster.

Further study needs to be conducted on (1) the interaction of demographic factors with the data features characterizing the clusters, and (2) the computational validation of the model.

The results support previous studies that highlighted the effect of the COVID-19 pandemic on mental health of people and the differences in response of people to stress. The finding that post-traumatic growth influences coping mechanism, and consequently, affects mental health condition can be a useful reference for further studies on pandemic-related coping characteristics of the Philippine population. Moreover, it can aid in the formulation of public health policies and intervention methods in times of a pandemic.

Appendix A

Environment and Source Codes

A.1 Environment

The Python notebook was run through Google Colab, and has the following dependencies:

- Numpy 1.22.4
- Pandas 1.5.3
- Scikit-learn 1.2.2
- Plotly 5.13.1
- Jupyter-dash 0.4.2
- Validclust 0.1.1
- Scipy 1.10.1
- Yellowbrick 1.5

A.2 Sections of the Python Notebook

Import Dataset

Imports the dataset into a pandas DafaFrame, and installs needed libraries not in Google Colab.

Get 5th Survey Data

Gets the dataset columns pertaining to the fifth survey.

Check if any column has missing (NaN) values

Checks for missing values among the features.

Preprocessing

Performs the following:

- Renames numerical codes into strings to make the values more readable.
- One-hot encodes categorical features.
- Drops selected features.
- Drops entries with unknown/blank personal or household income.
- Standardizes numerical features (makes mean=0 and standard deviation=1).
- Scales all features to the range [0, 1].

The Variables subsection indicates which features are categorical and numerical, and which features are related to demographics, mental health, COVID, and coping behavior.

Running the whole Preprocessing section generates all preprocessing versions.

Summary Statistics

Shows the summary statistics of each feature. For numerical features, it shows the mean, standard deviation, minimum, maximum, median, 25th, and 75th percentiles. For categorical features, it shows the counts of each feature value.

Data Visualization

Plots the feature distributions and correlations between features.

Model Construction

Performs clustering using several parameters and evaluates their performance. The subsection titles indicate the algorithm and the preprocessing version used for clustering.

Cluster Characterization

Characterizes the clusters formed from the Model Construction section. The clusters being characterized are indicated by subsection titles, which have the algorithm, the preprocessing version, and the parameters used for clustering.

References

References used to create the code.

A.3 Functions

build_evaluate_dbscan(eps, min_samples, dataset)

Takes in the pandas DataFrame dataset and clusters it using DBSCAN with parameters *eps* and *min_samples*. It also evaluates the clustering using silhouette score, Davies-Bouldin index, and Dunn index. The function returns the silhouette score of the clustering.

build_evaluate_kmeans(n_clusters, dataset)

Takes in the pandas DataFrame dataset and clusters it using K-means with parameter *n_clusters*. It also evaluates the clustering using silhouette score, Davies-Bouldin index, and Dunn index. The function returns the silhouette score of the clustering.

build_evaluate_agglomerative (dataset, n_clusters=2, distance_threshold=None)

Takes in the pandas DataFrame dataset and clusters it using agglomerative clustering with parameters *n_clusters* and *distance_threshold*. It also evaluates the clustering using silhouette score, Davies-Bouldin index, and Dunn index. The function returns the silhouette score of the clustering.

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